

SolWind

System User's Manual

Hourly solar, wind and battery simulation with day-ahead forecasting
for three Philippine sites

Version 1.0
October 2026

<https://habagat.streamlit.app>

Document Control

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Revision History

Version	Date	Description
1.0	October 2026	First issue. Covers all nine pages of the application, local installation, downloads, messages and troubleshooting.

Notice

All energy figures in SolWind are modeled from NASA POWER weather data using fixed engineering equations. They are not metered plant output. SolWind is intended for study, comparison and planning exercises. It must not be used as the sole basis for operating a power plant or dispatching a grid.

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1 General Information

1.1 System Overview

SolWind is a web application that simulates a 1 MW solar photovoltaic (PV) plant, a 1 MW wind turbine and a 2 MWh battery, hour by hour, at three Philippine sites from 1 January 2020 to 30 June 2026. It also scores day-ahead forecasts of that output and shows how forecast error changes an hourly grid-purchase plan.

With SolWind you can:

- replay any day at any site and watch output, demand and battery charge change hour by hour;
- resize the plant and battery and rerun every hour of the record;
- compare the three sites on energy, seasonality and wind;
- inspect day-ahead forecast accuracy for 2025 and January to June 2026;
- download hourly results as CSV files.

Site	Island group	Coordinates
Laoag, Ilocos Norte	Luzon	18.18° N, 120.53° E
Mactan, Cebu	Visayas	10.32° N, 123.98° E
General Santos City	Mindanao	6.06° N, 125.10° E

1.2 Intended Audience

This manual is for energy planners, engineers, analysts, instructors and students who use SolWind to study hybrid solar and wind generation. It assumes you can use a web browser. Section 3.2 assumes basic familiarity with installing software. No programming is needed to use the application.

1.3 Project References

Reference	Description
SolWind technical paper	“Day Ahead Forecasting of Modeled Solar and Wind Generation and Battery Planning in Three Philippine Locations” (IEEE conference format). Describes the equations, forecasting method and results in full.
Methodology page	The Methodology page inside SolWind (Section 4.9) lists the equations, fixed parameters, sources and limitations.
Source repository	https://github.com/RalphKGM/habagat-streamlit
NASA POWER	NASA Prediction Of Worldwide Energy Resources, hourly data. https://power.larc.nasa.gov

1.4 Authorized Use

SolWind is open to anyone with the address. No account is needed. The weather data comes from NASA POWER, which is free to use with attribution. Results may be quoted or reproduced if SolWind and NASA POWER are credited and the results are described as modeled, not measured.

1.5 Points of Contact

Report problems, questions and suggestions by opening an issue in the source repository: <https://github.com/RalphKGM/habagat-streamlit/issues>. Include the page name, the site and day selected, and what you expected to see.

1.6 Organization of the Manual

Section	Contents
1 General Information	Purpose, audience, references and conventions.
2 System Summary	Configuration, data flow, access and alternate modes of operation.
3 Getting Started	Opening the hosted system, installing it locally, the menu and shared controls.
4 Using the System	Each page in turn: purpose, controls, results and procedures.
5 Querying	Choosing a site, day, source and method to answer a specific question.
6 Reporting	Downloading results and the contents of each file.
7 Messages and Troubleshooting	On-screen messages, problems and fixes, frequent questions.
Appendices	Glossary, standard design parameters and index.

1.7 Document Conventions

Convention	Meaning	Example
Bold	A page, button, menu or control you click or set.	Select Reset to standard design .
<i>Italic</i>	An option within a control, or a term defined in the glossary.	Set Show to <i>Each hour</i> .
Monospace	Text you type, a file name or a command.	<code>streamlit run streamlit_app.py</code>
A > B	Open menu A, then select item B.	Forecasting > Rolling test
Note	Information that helps you avoid a mistake or misreading.	—

Times are Philippine Standard Time (PHT, UTC+8). Power is in megawatts (MW), energy in megawatt-hours (MWh).

1.8 Acronyms and Abbreviations

Term	Meaning
CSV	Comma-separated values (a spreadsheet-readable text file)
MAE	Mean absolute error
MW / MWh	Megawatt / megawatt-hour
NASA POWER	NASA Prediction Of Worldwide Energy Resources
PHT	Philippine Standard Time, UTC+8
PV	Photovoltaic (solar panel)
SOC	State of charge of the battery
XGBoost	Extreme Gradient Boosting, the machine-learning forecasting method

2 System Summary

2.1 System Configuration

SolWind runs in a web browser. It can be used from the hosted address or run on your own computer.

Component	Requirement
Browser	A current version of Chrome, Edge, Firefox or Safari, with JavaScript enabled. A screen at least 1280 pixels wide is recommended. Below 760 pixels the Map switches to a stacked phone layout.
Network	An internet connection for the hosted address. A local copy needs a connection only for its first start.
Local computer (optional)	macOS, Windows or Linux with Python 3.9 to 3.12 and about 500 MB of free disk space.
Python packages	streamlit 1.50 or later, plotly, pandas, numpy and pyarrow, installed automatically from <code>requirements.txt</code> .

2.2 Data Flows

Data moves through SolWind in five stages. All data ships with the application. Nothing is downloaded while you use it.

1. **Weather.** Hourly NASA POWER irradiance, temperature and wind for each site, 1 January 2020 to 30 June 2026 (56,952 hours per site).
2. **Generation.** Fixed equations turn weather into solar and wind output for the current design.
3. **Dispatch.** Each hour, generation serves the standard demand profile first. Surplus charges the battery, a shortfall is met from the battery, then from the grid.
4. **Forecasts.** Stored day-ahead forecasts from three methods, issued at 00:00 for each day, are compared with the generated output.
5. **Pages and downloads.** Results are shown as charts, tables and figures, and can be downloaded as CSV files.

Note: Forecasts are scored against the standard 1 MW + 1 MW design. Changing the design in System designer does not change the stored forecast results.

2.3 User Access Levels

SolWind has one access level. Every visitor can view every page and download every file. There are no accounts or passwords. Settings you choose, such as site, design and day, last only for your browser session and are not seen by other users.

2.4 Contingencies and Alternate Modes of Operation

Situation	Alternate mode
The hosted address is asleep	Select the wake-up button and wait about 30 seconds (Section 7.2).
The hosted address is unavailable	Run SolWind locally (Section 3.2). The local copy contains all data and works offline after its first start.
A page shows an error	Reload the page. If the error remains, restart the application or report it (Section 1.5).

3 Getting Started

3.1 Opening the Hosted System

1. Open a browser and go to `https://habagat.streamlit.app`.
2. If a wake-up screen appears, select the button and wait for the application to start.
3. The **Map** page opens. The first page can take a few seconds while the hourly results are calculated. Later pages open at once.

3.2 Installing and Running Locally

Using the start file (macOS or Windows)

1. Download the source code from `https://github.com/RalphKGM/habagat-streamlit` (**Code > Download ZIP**) and unzip it, or clone the repository.
2. On macOS, double-click `Run SolWind.command`. On Windows, double-click `Run SolWind.bat`.
3. Wait while the first start installs the required packages. This needs an internet connection and takes a few minutes.
4. Your browser opens SolWind at `http://localhost:8501`.

Note: If macOS says the file cannot be opened, right-click it, select **Open**, then confirm.

Using a terminal

```
python -m venv .venv
source .venv/bin/activate # Windows: .venv\Scripts\activate
pip install -r requirements.txt
streamlit run streamlit_app.py
```

3.3 System Menu

The bar across the top of every page is the system menu. Select a page name to open it. Pages are grouped as follows.

Menu group	Page	Purpose	Section
(main)	Map	All three sites on one day, hour by hour	4.1
(main)	Live plant	One site and day as an animated plant	4.2
(main)	System designer	Resize the plant and rerun every hour	4.3
(main)	Compare sites	The three sites side by side	4.4
Forecasting	Forecast accuracy	2025 day-ahead forecast errors	4.5
Forecasting	Grid planning	How forecasts change the grid-purchase plan	4.6

Menu group	Page	Purpose	Section
Forecasting	Rolling test	Each year forecast by a model trained on earlier years	4.7
Forecasting	2026 forecast	January to June 2026, computed against forecast	4.8
Reference	Methodology	Equations, parameters, sources and limits	4.9

Your site, system design and chosen day carry over as you move between pages.

3.4 Common Controls

Site picker

Selects Laoag, Mactan or General Santos. Pages that show one site have it near the top.

Day picker

Pages that show a single day use the same day picker: the date with < and > beside it.

Control	Action
< >	Step back or forward one day. The Left and Right arrow keys do the same.
The date	Opens a calendar. Use the month and year menus, or the arrows at the top, to change month, then click a day.
Type a date	Field at the bottom of the calendar. Accepts 2025-03-14, 14 Mar 2025, Mar 14 2025, 14/03/2025 (day first) or Mar 2025 (first day of the month). Press Enter.
Shortcuts	<i>Windiest, Sunniest, Calmest</i> (least energy) and <i>Random</i> , for the selected site.

With the calendar open, the Left and Right arrow keys move one day, Up and Down move one week, Enter opens the highlighted day and Esc closes the calendar.

Days range from 1 January 2020 to 30 June 2026. On Forecast accuracy and Grid planning the picker is limited to 2025, the year those forecasts were tested on.

Charts

Hover over a chart to read exact values. Drag across a chart to zoom in, and double-click to zoom back out. Click an item in a chart legend to hide or show that series.

3.5 Exiting the System

Close the browser tab. Nothing needs to be saved. To stop a local copy, close the terminal window it opened, or press Ctrl+C in that window.

4 Using the System

This section describes each page in menu order.

4.1 Map

The home page. A map of Luzon, the Visayas and Mindanao shows the three sites on one day, hour by hour.

Control	Location	Action
Key	Top left	Colors the regions by <i>Output</i> this hour, <i>Capacity factor</i> or <i>Wind share</i> .
Wind, Labels	Top right	Show or hide the moving wind particles and the site names. Particles follow each site's simulated hub-height wind.
Timeline	Bottom	Pick the day on the left, drag the yellow needle across the 24-hour chart, or select PLAY .
Regions and site tabs	Map, right panel	Hover a region to read its output, capacity factor and wind share. Click a region or site tab to select it.

The right-hand panel has two tabs. *This day* shows the hour's output, the share of demand met by renewables, battery charge, energy flows against demand, and averages for January 2020 to June 2026. *2025 forecast test* shows how each forecasting method scored at the selected site.

To compare the three sites on one day

1. Open **Map** and set the day in the timeline.
2. Select **PLAY**, or drag the needle, and set the key to *Capacity factor*.
3. Click each region and read its figures in the right-hand panel.

4.2 Live Plant

An animated plant for one site and day. The sky follows the hour, cloud cover follows the weather, the rotor speed follows wind output, and the dotted flows are scaled to MW.

Result	Meaning
Generated	Solar plus wind energy for the day, MWh.
Renewable share	Share of the day's demand met by solar, wind or the battery.
Bought from grid	Energy imported because generation and battery fell short, MWh.
Unused renewables	Surplus that could not be used or stored, MWh.
Peak hub wind	Highest wind speed at the 69 m hub that day, m/s.

Below these figures, an hourly chart shows solar, wind, demand and battery charge.

To see how a typhoon-season day played out

1. Open **Live plant** and choose a site.
2. Open the calendar and pick a day between July and October, or use the *Calkest* shortcut.
3. Watch the battery and the *From grid* flow as the day plays.

4.3 System Designer

Resize the plant and rerun every hour from January 2020 to June 2026 for the selected site.

Control	Range	Standard
Solar capacity · MW	0 to 3, steps of 0.25	1.0
Wind capacity · MW	0 to 3, steps of 0.25	1.0
Battery · MWh	0 to 8, steps of 0.5	2.0
Demand multiplier	0.5 to 1.5, steps of 0.1	1.0
Other PV losses (under Engineering assumptions)	0.05 to 0.25	0.14
PV temperature coefficient · /°C	−0.0060 to −0.0020	−0.0047
Wind-shear exponent	0.08 to 0.30	0.14
Wind net-output factor	0.70 to 1.00	0.90

Reset to standard design returns every control to its standard value.

Results show demand met by renewables, renewable energy, grid energy needed and unused renewables, each compared with the standard design. A yearly chart shows how demand was met. A month-by-hour chart, *Surplus by month and hour*, shows when the plant charges the battery (green) or draws on it (red).

Outage test

Simulates a grid outage. During the window, any shortfall becomes unmet demand.

1. Under **Outage test**, pick the day.
2. Set **Starts at** (0:00 to 23:00) and **Lasts · hours** (1 to 24).
3. Read *Demand served in outage*, *Unmet demand* and *First shortfall*. The chart shows the window with six hours either side.

To test a bigger battery

1. Move **Battery · MWh** from 2 to 4.
2. Read the change in *Demand met by renewables* and *Grid energy needed*.
3. Run the outage test on an evening to see whether the extra storage carries the load.

Note: Your design carries over to Map and Live plant. Select **Reset to standard design** to return to the figures used elsewhere in this manual.

4.4 Compare Sites

The three sites side by side for January 2020 to June 2026, with the same equipment at each: yearly energy by source, average output by month, wind roses (direction and speed at 69 m), and solar-wind complementarity. A complementarity value below zero means one source tends to fill in when the other is low.

4.5 Forecast Accuracy

Day-ahead forecasts issued at 00:00 for every day of 2025, a year held out from training. Three methods are compared: *XGBoost* (machine learning), *Previous day* (repeat yesterday) and *Past average* (the typical value for that month and hour).

Part of the page	What it shows
Leaderboard	Combined solar and wind error (MAE) for each method and site.
Source	Switches the charts between <i>Combined</i> , <i>Solar</i> and <i>Wind</i> .
Error by hours ahead and by month	Where each method does well or badly.
Single-day playback	One 2025 day for one method. Turn off Reveal the reference to see the forecast alone first. Turn on Show all three methods to overlay them.

To check a forecast on a specific day

1. Open **Forecast accuracy** and pick a 2025 day under *Single-day playback*.
2. Turn off **Reveal the reference**, play the day, then turn it back on to compare.
3. Open **Grid planning** with the same day to see what that forecast did to the grid plan.

4.6 Grid Planning

Each forecast sets an hourly grid-purchase plan at midnight. The real day is then replayed. *Adjustment* is the extra energy bought plus planned energy that was not needed. The page totals adjustment for 2025 by method. Under *Day replay*, pick a site, method and day to watch the plan (fixed at 00:00) against actual output.

4.7 Rolling Test

The forecast rerun for 2024, 2025 and January to June 2026, each year predicted by a model that only saw earlier years.

- The fold grid shows which years were used to fit, select and test each model.
- *Every day, at midnight*: a calendar colors each day green where XGBoost beat the previous-day forecast and red where it lost. Use **Test** to choose the year. Click a day to open its 24-hour forecast.

- *Is the gap real?*: 95% intervals for the error difference from a seven-day block bootstrap (2,000 resamples). An interval left of zero favors XGBoost.
- *Shorter history?*: the same 2026 hours forecast with and without 2020 in training.
- *Other models*: Random Forest and Ridge (a linear model) put through the same steps on the same hours, with each test year's error beside XGBoost and the two simple methods, and whether the XGBoost and Random Forest difference is real.

A link at the top of the page opens 2026 forecast.

4.8 2026 Forecast

Shows that the January to June 2026 forecasts came from models trained only on 2020 to 2025, and how they compare with the output computed from 2026 weather.

Step	Data	Hours per site and source	Purpose
1 Fit	2020–2024	43,824	Two candidate settings are trained.
2 Select	2025	8,760	The better setting is kept.
3 Refit	2020–2025	52,584	Retrained. The last hour is 31 Dec 2025, 23:00.
4 Forecast	Jan–Jun 2026	4,344	Issued at 00:00 each day from the day before.
5 Score	Jan–Jun 2026	4,344	Scored once against computed output.

- *Computed vs forecast*: set **Show** to *Daily energy* for each 2026 day, or *Each hour* for all 4,344 hours against a perfect-forecast line plus one week in detail. A monthly table lists energy and error for all three methods.
- *Recompute it*: SolWind reruns its own equations on 2026 weather and compares every hour with the values the forecasts were scored against. The largest difference is under 0.000001 MW.
- *Download*: every 2026 hour at all three sites (Section 6.2).

To show that 2026 was forecast from earlier years only

1. Open **Forecasting > 2026 forecast** and read the five steps.
2. Set **Show** to *Each hour* to compare computed and forecast output.
3. Read *Largest difference* under *Recompute it*.
4. Select **Download 2026 computed vs forecast (CSV)** to keep the hourly evidence.

4.9 Methodology

Reference material in tabs: *Solar PV*, *Wind turbine*, *Battery* and *Forecasting*, each with its equations. Below the tabs are the fixed parameters, a comparison with related IEEE studies, the data sources and the limitations of the model.

5 Querying

SolWind has no search box. You query it by choosing a page and setting its site, day, source and method. The table lists common questions and where to answer them.

Question	Page	Settings
How much did a site generate on a given day?	Live plant	Site, day
Which site has the best capacity factor at a given hour?	Map	Day, key = <i>Capacity factor</i> , needle on the hour
What was the windiest day at a site?	Live plant or Map	Day picker shortcut <i>Windiest</i>
How much grid energy would a 4 MWh battery save?	System designer	Battery = 4, compare with standard
Can the plant ride through an evening outage?	System designer	Outage test: day, start, length
Which forecasting method is most accurate?	Forecast accuracy	Leaderboard, source
In which months do forecasts miss most?	Forecast accuracy	Site, source, error by month
How much does forecast error cost the grid plan?	Grid planning	Site, method
Is XGBoost's advantage real?	Rolling test	<i>Is the gap real?</i>
Was 2026 forecast without seeing 2026?	2026 forecast	Five steps, <i>Recompute it</i>

6 Reporting

SolWind reports results on screen and as CSV files that open in any spreadsheet program. Times in every file are PHT.

6.1 Hourly Results for a Design

1. Open **System designer** and set the site and design.
2. Open **Hourly results table and download** at the bottom of the page. The table shows the hours of the outage-test day.
3. Select **Download every hour, 2020 to June 2026, for this design (CSV)**.

The file is named `solwind_<site>_2020_2026.csv` and has one row per hour (56,952 rows).

Column	Unit	Meaning
timestamp_pht	—	Start of the hour, PHT
solar_mw	MW	Solar output
wind_mw	MW	Wind output
combined_mw	MW	Solar plus wind output
demand_mw	MW	Demand, after the demand multiplier
battery_soc_end_mwh	MWh	Battery charge at the end of the hour
grid_import_mwh	MWh	Energy bought from the grid
curtailed_renewable_mwh	MWh	Surplus that could not be used or stored

6.2 2026 Computed vs Forecast

1. Open **Forecasting > 2026 forecast**.
2. Under *Download*, select **Download 2026 computed vs forecast (CSV)**.

The file is named `solwind_2026_computed_vs_forecast.csv` and holds every hour from 1 January to 30 June 2026 at all three sites.

Column	Unit	Meaning
location	—	Site name
target_timestamp_pht	—	The hour being forecast, PHT
issued_at_pht	—	When the forecast was issued (00:00 of the same day)
computed_mw	MW	Output computed from that hour's weather (the reference)
xgboost_forecast_mw	MW	XGBoost forecast
previous_day_forecast_mw	MW	Previous-day forecast
training_climatology_forecast_mw	MW	Past-average forecast

7 Messages and Troubleshooting

7.1 On-Screen Messages

Message or sign	Meaning	Action
Wake-up screen (“This app has gone to sleep”)	The hosted copy sleeps after a period without visitors.	Select the button and wait about 30 seconds.
Running indicator in the top-right corner	The page is calculating.	Wait for it to finish.
Typed date turns red	The date was not recognized or is outside 1 January 2020 to 30 June 2026.	Use a listed format, for example 14 Mar 2025.
Red error box on a page	The page failed to load.	Reload the page. If it remains, see Section 7.2.

7.2 Problems and Fixes

Find the problem in the left column and follow the fix.

Problem	Fix
The first page takes a few seconds	The equations run once over 170,856 site-hours and are then cached. Later pages are instant.
The map or calendar looks cut off	Widen the browser window. Below 760 pixels the map switches to a stacked phone layout.
Arrow keys do nothing	Click inside the map or day picker first so it has keyboard focus.
Numbers differ from those in this manual	Your System designer settings carry across pages. Select Reset to standard design .
The local start file closes at once	Check that Python 3.9 to 3.12 is installed and on the system path, then run the terminal commands in Section 3.2.
A page shows an error after an update	On the hosted copy, reload the page. On a local copy, stop it and start it again.
A download does not start	Allow downloads for the site in your browser settings, then select the button again.

7.3 Frequently Asked Questions

Is this real plant data? No. Output is modeled from NASA POWER weather data for each site's grid cell. No plant was metered, and demand is one standard daily profile used at every site.

Why only next-day forecasts? Grid purchases are scheduled the day before, and weather becomes much less predictable after a day or two.

Do I have to run forecasts myself? No. Forecasts are stored for every day. The pickers only choose which day to inspect.

Why does the record stop at 30 June 2026? NASA had not released complete solar data for later months when the data was collected.

Can SolWind forecast 2027? No. A full-year outlook needs a separate long-term resource study, which SolWind does not provide.

What is outside the model? Voltage, frequency, equipment failures, typhoon damage and costs. SolWind covers hourly energy only.

Appendix A Glossary

Term	Meaning
Adjustment	Extra energy bought plus planned energy that was not needed, after the real day is known. Lower means the plan was closer to reality.
Battery charge (SOC)	Energy stored, kept between 10% and 90% of capacity. Charging and discharging are each 95% efficient.
Capacity factor	Average output divided by installed capacity. 20% means the plant produced a fifth of its maximum possible energy.
Computed output	The modeled generation the forecasts are scored against, calculated from the actual weather of that hour. Also called the reference.
Day-ahead forecast	A prediction of the next 24 hours, made at 00:00 using only information available then.
Demand met by renewables	Share of demand served by solar, wind or the battery rather than the grid.
Fold	One run of the rolling test: years to fit the model, a year to choose its settings, and a later period to test it.
Grid energy needed	Energy bought from the grid when solar, wind and the battery fall short.
Hub-height wind	Wind speed at the turbine's 69 m hub, estimated from NASA's 50 m wind.
MAE	Mean absolute error: the average size of the forecast miss, in MW. Lower is better.
MW / MWh	Power at an instant / energy over time. 1 MW for one hour is 1 MWh.
Past average	A simple forecast using the typical output for that month and hour in the training years.
Previous day	A simple forecast that repeats yesterday's hourly output.
Unused renewables	Surplus that could not be used or stored because the battery was full.
XGBoost	The machine-learning forecast: a gradient-boosted tree model that learns from the previous day's output and the calendar.

Appendix B Standard Design Parameters

These values are used whenever the standard design is selected. They can be read in full on the Methodology page.

Parameter	Value
PV capacity (DC / AC limit)	1.0 MW / 1.0 MW
PV tilt and azimuth	10°, facing south (180°)
Ground albedo	0.20
PV temperature coefficient	−0.0047 per °C, reference 25 °C
Other PV losses	14%
Inverter efficiency	96%
Wind turbine	EWT DW61, 1 MW, 69 m hub height
Wind source height	50 m (NASA POWER)
Wind-shear exponent	0.14
Reference air density	1.225 kg/m ³
Wind net-output factor	0.90
Battery	2 MWh, 10% to 90% usable, 95% charge and 95% discharge efficiency
Demand	Standard daily profile: 0.25 MW overnight, rising to 0.65 MW in the evening

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